

NIELS BUGEL

Software Engineer @ CERN

Software developer with a passion for high-performance computing, distributed systems and software architecture. I like writing documentation and am a strong believer in simplicity, testing and automation. I may not know the solution to every question or problem by heart, but I can certainly find out.

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WORK EXPERIENCE

Software Engineer **CERN** **May 2024 – Current**

- C++ developer and operations support for CTA: a large-scale distributed tape storage system managing over 1 exabyte of data.
- Integrated OpenTelemetry Metrics into CTA, collaborated with the monitoring team at CERN to efficiently use the provided infrastructure and defined metrics to assess system health and performance
- Rewrote the CTA GitLab CI pipeline and designed a new system test framework based on pytest, reducing merge pipeline times from 50 min to 15 min
- Implemented a C++ library for a more robust and consistent application setup
- Redesigned the CTA CI Kubernetes orchestration using Helm to improve modularity and testing flexibility
- Started a team-wide initiative to rewrite, improve and keep up-to-date the CTA documentation
- Took various other initiatives to reduce technical debt and introduce development best practices

Backend Software Engineer **Dataqueue** **Feb 2024 – Apr 2024**

- Implemented the backend infrastructure for a real-time AI-based call analytics platform using Python/TypeScript microservices, Kafka, WebSockets, and Kubernetes (GKE, Helm)

Teaching Assistant **University of Groningen** **Feb 2019 - July 2023**

- **BSc courses:** Object-Oriented Programming (x5; Coordinator), Advanced Object-Oriented Programming (x4; Coordinator), Signals and Systems (x2), Algorithms and Data Structures (x2), Advanced Algorithms and Data Structures (x2), Parallel Computing (x2), Computer Graphics, Operating Systems
- **MSc courses:** Advanced Computer Graphics, Advanced Parallel Programming, Image Processing (x2)
- **Student Mentor**

Numerus Fixus Coordinator CS **University of Groningen** **Oct 2019 – July 2023**

- Designed, wrote and coordinated the selection procedure taken by over 400 students yearly

Full Stack Developer **University of Groningen** **Oct 2020 – Aug 2022**

- Designed and built MATIG, a web application that automated the matching procedure for several studies at the Faculty of Science and Engineering (React, Express.js, MongoDB, Redis; Kubernetes)

LANGUAGES, TECHNOLOGIES AND SKILLS

- C++ (17, 20), C, Python, TypeScript, JavaScript, Java
- Kubernetes, GitLab CI, OpenTelemetry Metrics, Helm, Docker, SQL, Bash, Git

EDUCATION

Groningen, NL **University of Groningen** **Sept 2017 – July 2023**

- MSc. in Computing Science, Data Science & Systems Complexity, cum laude (GPA: 9.0/10)
- BSc. in Computing Science, cum laude (GPA: 8.8/10)

PUBLICATIONS

- N. Bugel, L. Romani, & J. Kosinka (2025). CONIS: A C++ Package for Conic-Preserving Interpolatory Subdivision. *Journal of Approximation Software*, 2(2). <https://doi.org/10.13135/3103-1935/11442>
- N. Bugel, L. Romani, & J. Kosinka (2024). A point-normal interpolatory subdivision scheme preserving conics. *Computer Aided Geometric Design*, Vol. 111. <https://doi.org/10.1016/j.cagd.2024.102347>

PROJECTS

NITRO

GitHub

- A node editor that allows for building complex non-destructive image processing pipelines (C++, Qt, OpenCV, and OpenGL)
- Comes with documentation: <https://www.nielsbugel.dev/nitro/>

CONIS

GitHub

- Application and framework for the conic subdivision scheme in <https://doi.org/10.1016/j.cagd.2024.102347> (C++, Qt, OpenGL, Eigen and GoogleTest)

Distributed GPU Convolution

GitHub

- A parallel implementation of generalized convolution operators for large image data sets (C, CUDA, pthreads, MPI)

For a more complete list of projects, see my personal website.

ADDITIONAL EXPERIENCE AND AWARDS

- Contributed to open-source projects such as opentelemetry-cpp and reflect-cpp
- Successfully completed the CERN School of Computing 2024
- CERN Guide for the Data Centre Visit Point and Science Gateway
- CERN Trainer for the Self-Rescue Mask course
- Speaker, Inverted CERN School of Computing — Writing Maintainable Code (2025)
- Speaker, Does It Compute? — Scalable Tape Software (2025)
- Initialised a collaboration initiative between CERN and the University of Groningen
- 4x best presentation award for various CS courses (*Introduction to Data Science*, *Student Colloquium (x2)*, *Information Systems*)
- 3x student member of interview committees (*Teachers for the Computing Science Programme*, *Tenure Track Assistant Professor in Visual Computing*, *Tenure Track Assistant Professor in Embedded Systems*)